

6.Introduction of Data Communication and Networking

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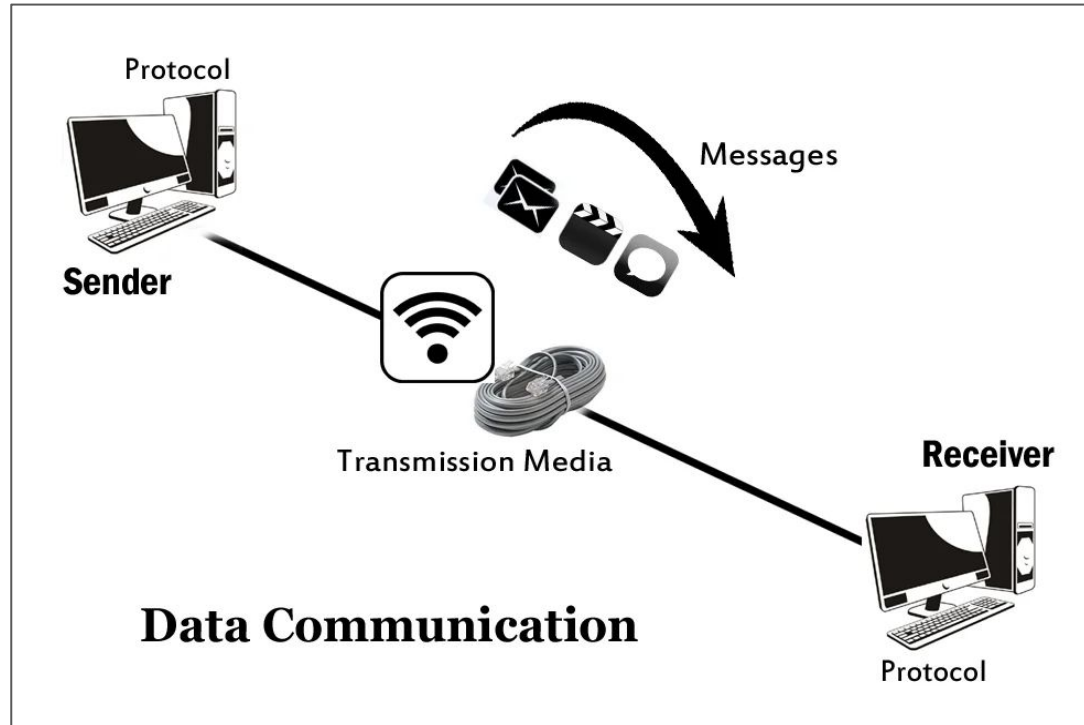
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Point-to-Point, Bus, Star, Ring, Mesh, Tree, Hybrid
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1. Data Communication Introduction

- ❑ Data communication is the process of transferring data between two or more devices through a transmission medium such as cable, fiber optic, or wireless signals.
- ❑ Main components of data communication
 - **Sender:** The device that sends the data
 - **Receiver:** The device that receives the data
 - **Message:** The actual data or information being sent
 - **Transmission medium:** The path used to send the data
 - **Protocol:** The rules that control communication
- ❑ Characteristics of effective data communication
 - **Delivery:** Data must reach the correct destination
 - **Accuracy:** Data should arrive without errors
 - **Timeliness:** Data should arrive on time, especially for voice and video

1. Data Communication Introduction



- ❑ **Example:** When we send a message on WhatsApp, our phone is the sender, the other person's phone is the receiver, the text is the message, and the internet is the transmission medium.

2. Computer Networking Introduction

- ❑ A computer network is a group of two or more computers and devices connected together so they can share data, resources, and services. The connection may be through cables or wireless signals.
- ❑ Main purposes of a computer network
 - Sharing files and information
 - Sharing hardware like printers and scanners
 - Sharing internet access
 - Enabling communication through email, chat, and video calls
 - Allowing centralized data storage and management
- ❑ **Example:** In a college computer lab, all computers may be connected to a single network. Students can print documents on one shared printer and access the internet through the same connection.

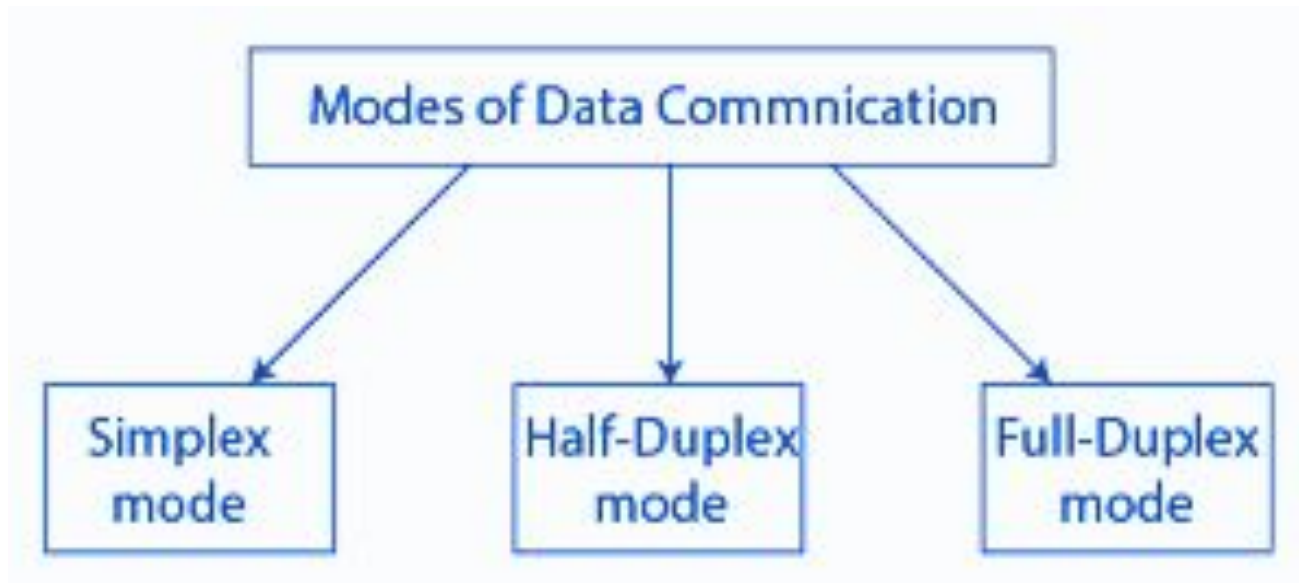
2. Computer Networking Introduction

❑ **Relation Between Data Communication and Networking:**

- A computer network is the system, while data communication is the process of sending information through that system.
- In short, networks allow devices to connect, and data communication allows those devices to exchange information.
- Data communication refers to the exchange of data between devices using a transmission medium and protocols, whereas computer networking is the interconnection of multiple devices to share resources and communicate.
- Data communication is a part of networking, focusing on the transmission process, while networking provides the overall structure and system for communication.
- **Focus:**
 - ***Data Communiation:*** Transfer of data (how data moves)
 - ***Networking:*** Connection of devices (how systems are linked)

3. Data Transmission(Communication) Modes

- ❖ Modes of data transmission describe the direction of data flow between two communicating devices. There are three fundamental modes:



3. Data Transmission(Communication) Modes

1. Simplex Mode

Definition:

Data flows in only one direction from sender to receiver. There is no feedback.

Characteristics

- One-way communication
- No reverse transmission possible
- Simple and low-cost

Examples

- Keyboard → Computer
- Television broadcast
- Radio transmission

2. Half-Duplex Mode

Definition:

Data can flow in both directions, but not at the same time.

Characteristics

- Two-way communication
- Only one device transmits at a time
- Requires coordination (turn-taking)

Examples

- Walkie-talkie
- Two-way radio communication

3. Full-Duplex Mode

Definition:

Data flows in both directions simultaneously.

Characteristics

- Two-way communication at the same time
- Faster and more efficient
- More complex and costly

Examples

- Telephone calls
- Video conferencing
- Modern internet communication

3. Data Transmission(Communication) Modes



Mainframe



Direction of data



Monitor

SIMPLEX



workstation



Direction of data at time 1



Direction of data at time 2



workstation

HALF DUPLEX



workstation



Direction of data all the time



workstation

FULL DUPLEX

4. Computer Network Types

Computer networks can be classified in several ways. The most common classifications are based on **geographical area**, **connection type**, **ownership/access control**, and **internetwork**.

Basis	Categories
Geographical area	PAN, LAN, MAN, WAN, GAN
Connection medium	Wired (copper/fiber) / Wireless (radio/IR)
Ownership & access	Private, Public, VPN (a logical overlay)
Interconnection	Internetwork (e.g., Internet, Intranet, Extranet)

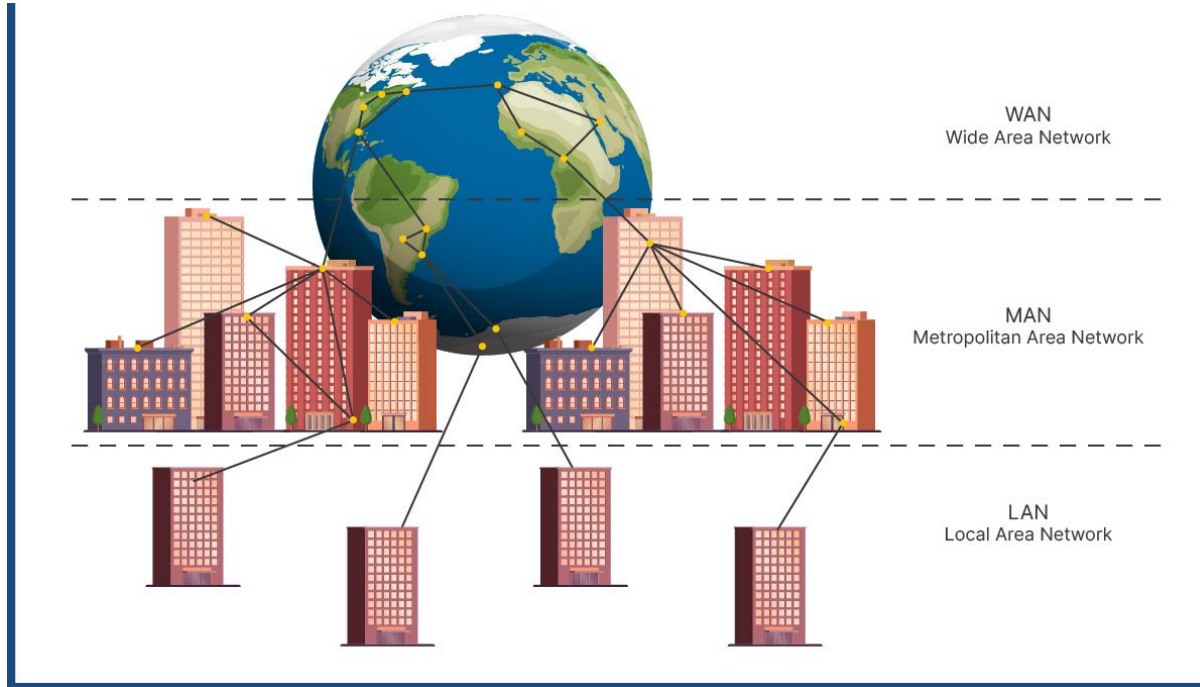
4. Computer Network Types

1. Based on Geographical Area (Scale)

Type	Full Form	Coverage Range	Examples
LAN	Local Area Network	Single building, office, home, or campus (up to ~1 km)	Ethernet in an office, Wi-Fi at home
MAN	Metropolitan Area Network	A city or large campus (5–50 km)	Cable TV network, city-wide Wi-Fi, university inter-building links
WAN	Wide Area Network	Country, continent, or global (100+ km)	Internet, banking network, corporate WAN connecting offices worldwide

4. Computer Network Types

1. Based on Geographical Area (Scale)



4. Computer Network Types

2. Based on Wired / Wireless Connection

Type	Medium	Characteristics	Speed & Reliability	Example
Wired Network	Physical cables (copper, fiber)	High reliability, low interference, secure	Fast (1 Gbps – 100+ Gbps)	Office LAN with Ethernet, fiber backbone
Wireless Network	Radio waves, infrared, microwave	Mobility, easy deployment, but susceptible to interference	Slower (typically 100 Mbps – 1 Gbps for Wi-Fi 6/7)	Wi-Fi, 4G/5G, Bluetooth, satellite

Hybrid networks (most real-world networks) combine both, e.g., smartphones connected via Wi-Fi (wireless) to a router that uses wired Ethernet/fiber to the ISP.

4. Computer Network Types

3. Based on Ownership and Access Control

Type	Ownership	Access	Security & Control	Example
Private Network	Owned by an organization / individual	Restricted to authorized users	High (firewall, encryption, authentication)	Home Wi-Fi, corporate LAN, VPN
Public Network	Owned by a service provider / telecom	Open to general public (with or without fee)	Low (shared, less trusted)	Public Wi-Fi at a café, telephone network
VPN (Virtual Private Network)	Uses public infrastructure (e.g., internet) but appears private	Authorized remote users	High (tunneling + encryption)	Employee working from home connecting to office network

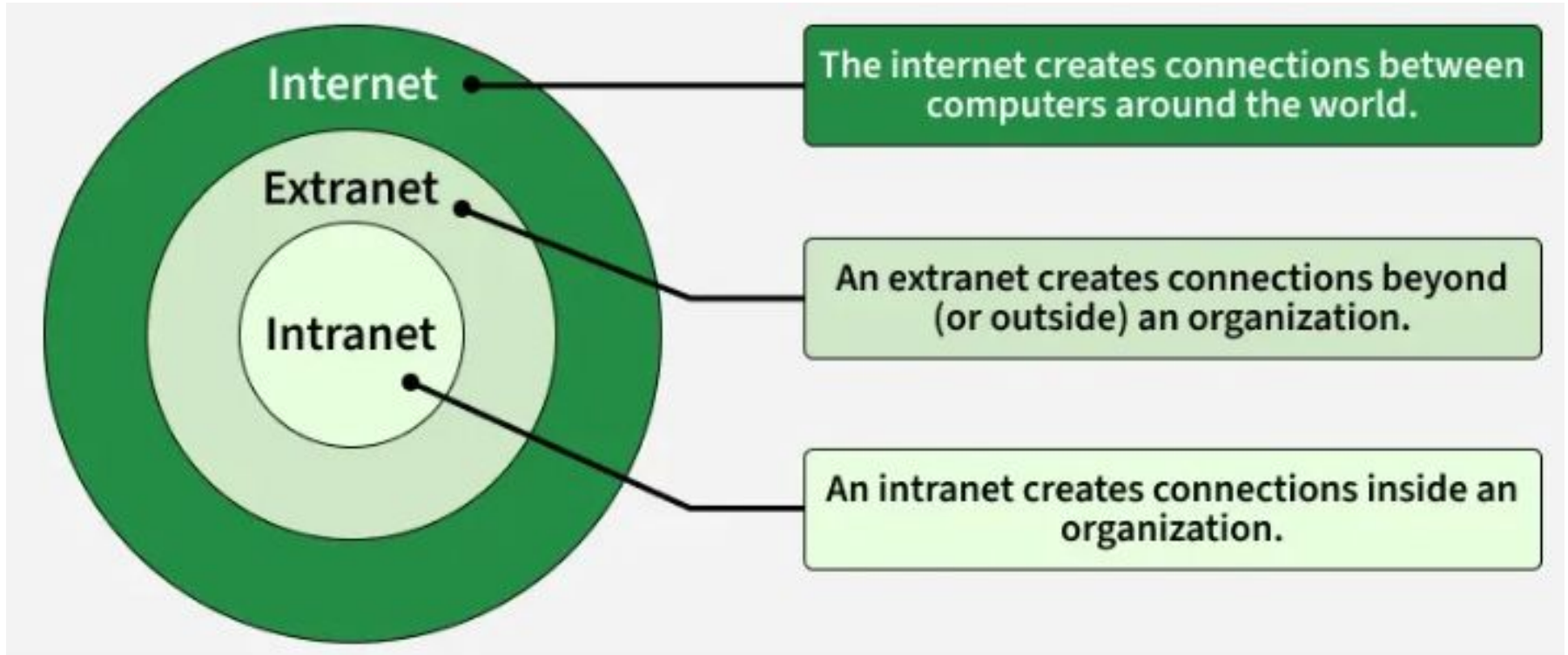
4. Computer Network Types

4. Internetworking

- ❑ Internetworking is the practice of linking together two or more computer networks using devices such as routers, gateways, or switches.
- ❑ The goal is to allow data to move between networks regardless of their underlying architecture.
- ❑ The most prominent example of internetworking is the Internet, which connects millions of smaller networks worldwide.
- ❑ Internetworking is a foundational concept in computer networking—it refers to the process of connecting multiple independent networks so they function as a single, larger network. This interconnected system enables seamless communication across different physical and logical network boundaries.
- ❑ Types of Internetworking:
 - i. Intranet
 - ii. Extranet
 - iii. Internet

4. Computer Network Types

4. Internetworking



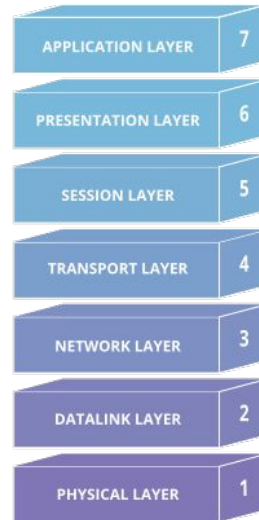
5. Computer Network Models

- ❑ **Network Model:** A network model is a **conceptual framework** that defines how data is transmitted, organized, and communicated across a network. It breaks down the complex process of network communication into **structured layers**, where each layer has specific responsibilities and interacts with the layers above and below it.
- ❑ **Need of Network Model:**
 - ❑ Simplify complex networking processes
 - ❑ Enable interoperability between different systems/vendors
 - ❑ Allow modular design — each layer can be developed independently
 - ❑ Standardize communication protocols
- ❑ The **two most important network models** are the **OSI Model** and the **TCP/IP Model**.

****A protocol** is a standardized, formal set of rules, conventions, or procedures governing how data is transmitted between network devices or how actions are performed in formal situations. **

5.1. Computer Network Models:OSI

- ❑ **OSI Model (Open Systems Interconnection):** Developed by ISO (International Organization for Standardization) in 1984. It divides **data transmission into seven distinct layers**, enabling specialists to troubleshoot, develop, and understand complex network interactions. The 7 layers are:



- ❑ Data flows from Layer 7 down to Layer 1 (**sending**) and
- ❑ Data flows from Layer 1 up to Layer 7 (**receiving**).

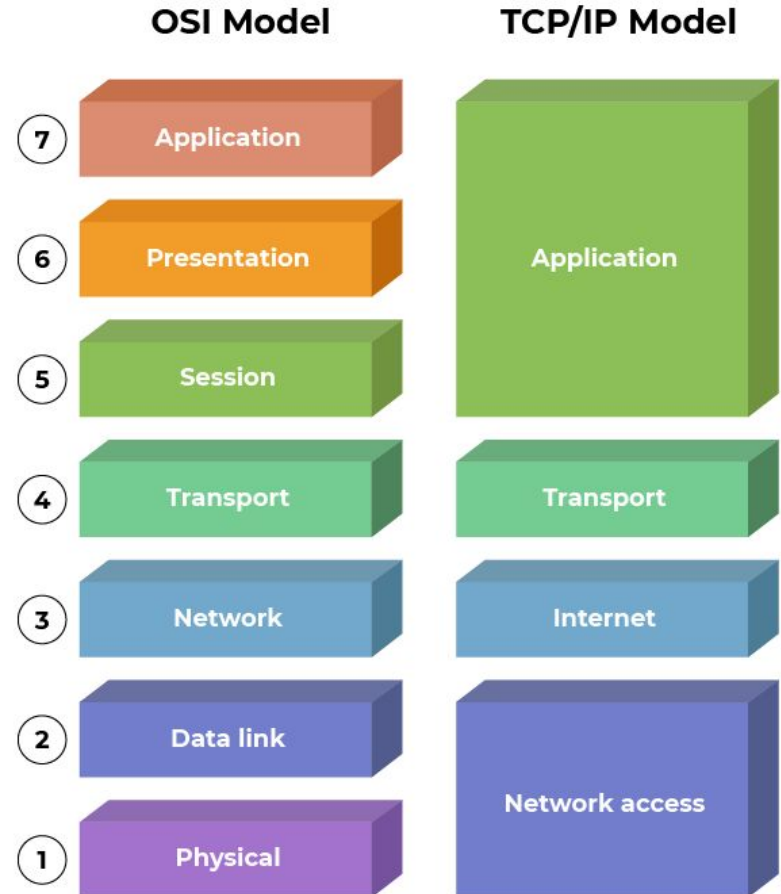
5.1. Computer Network Models:OSI Model

❑ The Seven Layers of the OSI Model

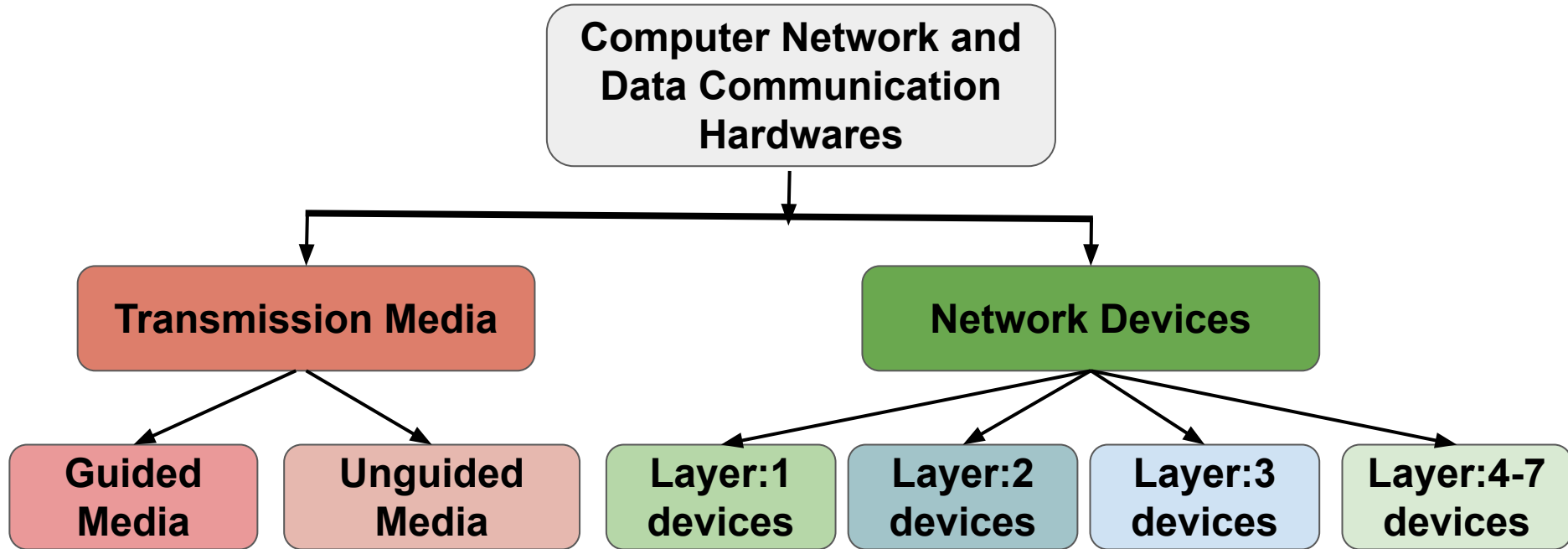
1. [Physical](#) (Layer 1): Handles the physical transmission of raw bitstreams over hardware (cables, hubs).
2. [Data Link](#) (Layer 2): Ensures reliable node-to-node transfer, framing data, and handling error detection/MAC addressing.
3. [Network](#) (Layer 3): Manages routing of packets across different networks using logical addressing (e.g., IP).
4. [Transport](#) (Layer 4): Enables end-to-end communication, flow control, and segmentation (e.g., TCP, UDP).
5. [Session](#) (Layer 5): Manages, maintains, and terminates connections between applications on different hosts.
6. [Presentation](#) (Layer 6): Translates, encrypts, and compresses data to ensure it is usable by the application layer.
7. [Application](#) (Layer 7): Serves as the user interface, allowing software applications to access network services.

5.2. Computer Network Models:TCP/IP

- ❑ The TCP/IP model is the practical model used in real-world internet communication.
- ❑ Developed by the U.S. Department of Defense (DARPA).
- ❑ Practical and implemented model, the backbone of the Internet
- ❑ It has 4 layers:
 1. Network Access
 2. Internet
 3. Transport
 4. Application



6. Computer Network and Data Communication Hardwares

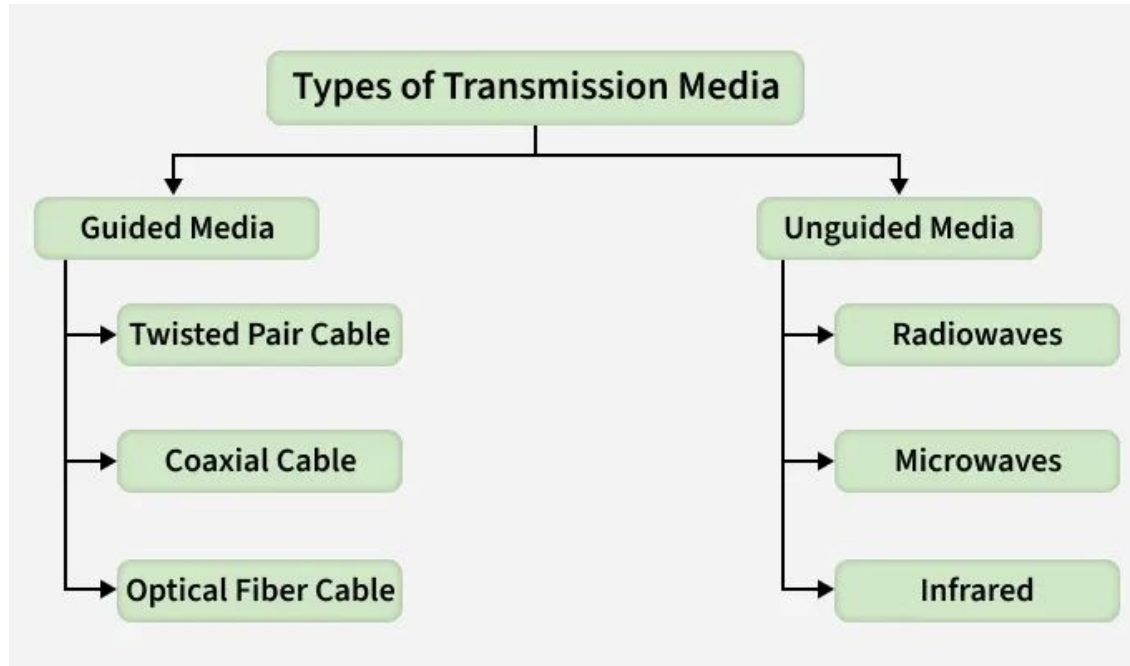


Transmission media defines *how* data physically travels (wire or air), while **network devices** control *where* data is directed and how efficiently it reaches its destination.

6.1. Transmission Media

- ❑ Transmission media refers to the physical or wireless communication channel used to carry data signals from one device to another within a computer network. It forms the fundamental pathway through which information is transmitted, ensuring connectivity between networked devices.
- ❑ The selection of a transmission medium depends on factors such as:
 - Transmission distance
 - Data transfer speed (bandwidth)
 - Susceptibility to interference and noise
 - Cost and installation requirements
- ❑ Based on the nature of the transmission path, transmission media are broadly classified into two main types:

6.1. Transmission Media



Guided media uses a **physical path (cables)** to transmit signals. The data signal is **directed and confined** within the medium.

Unguided media transmits signals through **air, vacuum, or space** without physical cables. Signals propagate as **electromagnetic waves**.

6.1.1. Transmission Media: Guided Media (Wired)

A. Twisted Pair Cable

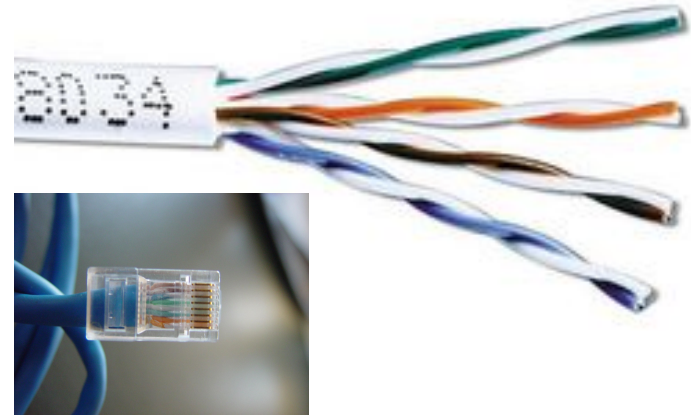
The most common and cost-effective LAN cabling. It consists of pairs of insulated copper wires twisted together to reduce electromagnetic interference (EMI).

- **Unshielded Twisted Pair (UTP):** No extra shielding; relies on twisting to cancel noise. It is cheap, flexible, and easy to install, making it common for Ethernet and telephone lines (CAT5e, CAT6).
- **Shielded Twisted Pair (STP):** Has a metallic foil or braided shield around pairs or the whole cable. It offers better noise resistance, making it suitable for industrial environments with high interference.
- **Connector:** RJ45.

Shielded twisted pair (STP)



Unshielded twisted pair (UTP)

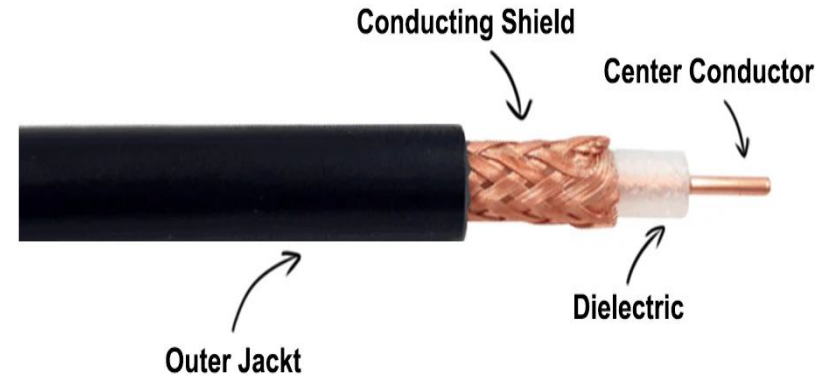


6.1.1. Transmission Media: Guided Media (Wired)

B. Coaxial Cable

A single copper conductor at the center, surrounded by a plastic insulation layer, a braided metal shield, and an outer jacket. This construction provides excellent resistance to interference.

- **Structure:** Carries high-frequency signals.
- **Legacy Use:** Traditional Ethernet (10BASE5, 10BASE2).
- **Modern Use:** Widely used for cable television (CATV) and broadband internet connections.



Coaxial Cable Construction

6.1.1. Transmission Media: Guided Media (Wired)

C. Fiber Optic Cable

Transmits data as pulses of light through glass or plastic fibers. It offers the highest performance and is immune to electromagnetic interference.

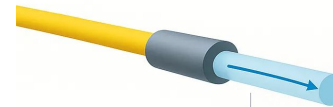
- **Single-Mode Fiber (SMF):** Extremely thin core (8-10 microns); uses laser light. It supports very long distances (up to 100 km) with high bandwidth, making it the backbone for telecommunications and WANs.
- **Multi-Mode Fiber (MMF):** Thicker core (50-62.5 microns); uses LED light. Light bounces along multiple paths, causing modal dispersion, which limits distance (up to ~550m). It's cheaper and commonly used for building backbones and data centers.



Single mode Fiber

Multi mode Fiber

SINGLE MODE



9 microns



long distance
up to 100km+

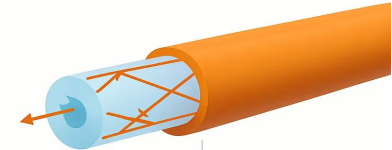


higher bandwidth



laser

MULTIMODE



50-62,5 microns



shorter distance
up to 2km



lower bandwidth



LED

❑ Comparison among twisted pair cable, coaxial cable, fibre optic cable

Feature	Twisted Pair Cable	Coaxial Cable	Fiber Optic Cable
Core Material	Insulated Copper wires	Solid Copper conductor	Glass or Plastic fiber
Signal Type	Electrical (electrons)	Electrical (electrons)	Light (photons)
Noise Immunity	Low (susceptible to EMI)	Medium (shielded)	Extremely High (immune to EMI)
Attenuation	High (signal dies quickly)	Lower than Twisted Pair	Very Low (signal travels far)
Bandwidth	Low to Moderate (up to 10 Gbps)	Moderate (historically up to 10 Mbps, modern DOCSIS higher)	Very High (Tbps theoretically)

Feature	Twisted Pair Cable	Coaxial Cable	Fiber Optic Cable
Max Segment Length	~100 meters	~500 meters	Multi-mode: ~550m to 2km Single-mode: up to 100km+
Security	Low (easy to tap, radiates signal)	Medium (harder to tap than UTP)	Very High (almost impossible to tap without detection)
Size & Weight	Thin, very flexible	Bulky, less flexible	Thin, lightweight, brittle
Installation Cost	Inexpensive, easy	Moderate	Expensive, requires specialist tools
Typical Connector	RJ45	BNC, F-Type	SC, LC, ST

6.1.2. Transmission Media: Unguided Media(Wireless)

Unguided media, also known as wireless or Unbounded transmission media, uses electromagnetic waves to transmit data without any physical medium. Signals propagate through free space such as air or vacuum. The main types of unguided media are radio waves, microwaves, and infrared waves.

Features:

- Signals propagate through air or free space
- Less secure due to broadcast nature
- Suitable for long-distance communication

6.1.2. Transmission Media: Unguided Media(Wireless): Radio Waves

Radio waves are a type of electromagnetic wave that can easily be generated and can propagate through buildings and other obstacles. They do not require line-of-sight between transmitting and receiving antennas, making them highly suitable for broadcast communication and wireless data transmission.

Frequency Range: 3 kHz – 1GHz

Applications:

AM and FM radio broadcasting, television transmission, cordless phones, and wireless communication.

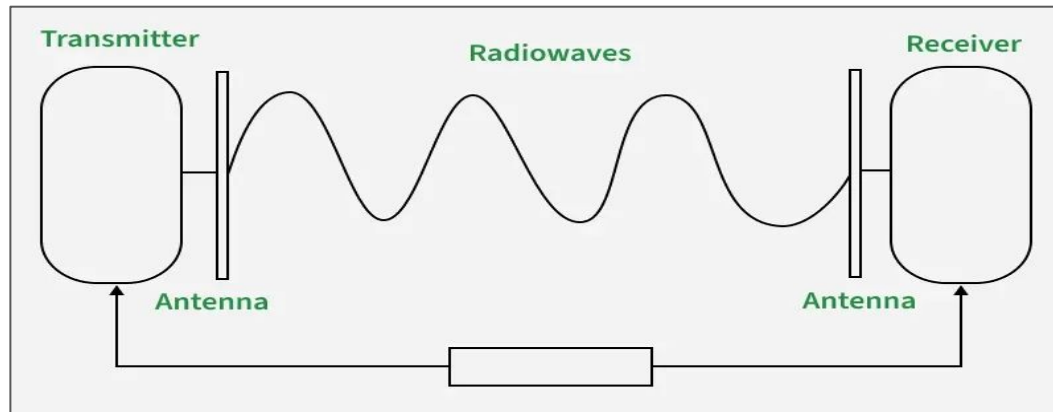
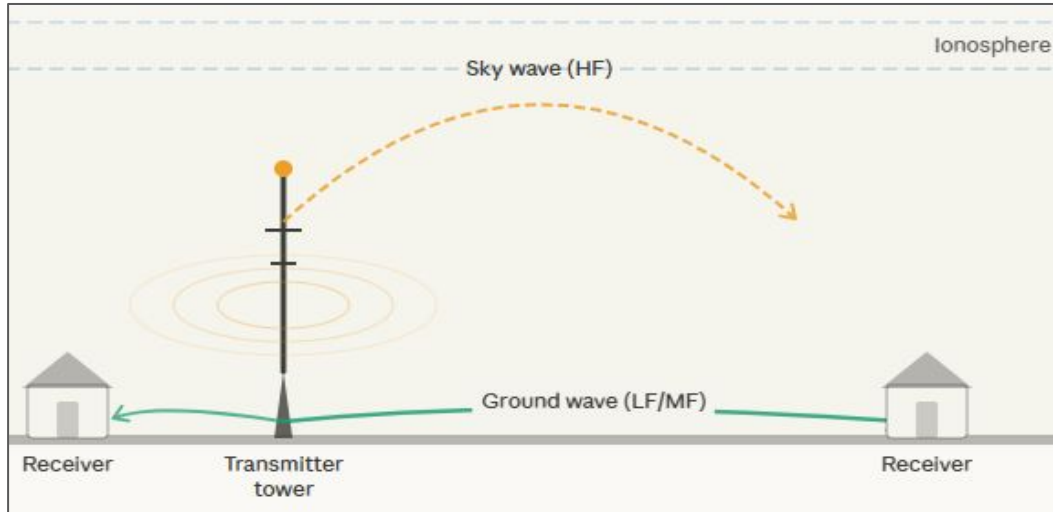
Types of Radio Waves:

- **Shortwave:** AM radio broadcasting
- **VHF (Very High Frequency):** FM radio and television
- **UHF (Ultra High Frequency):** Television and mobile communication

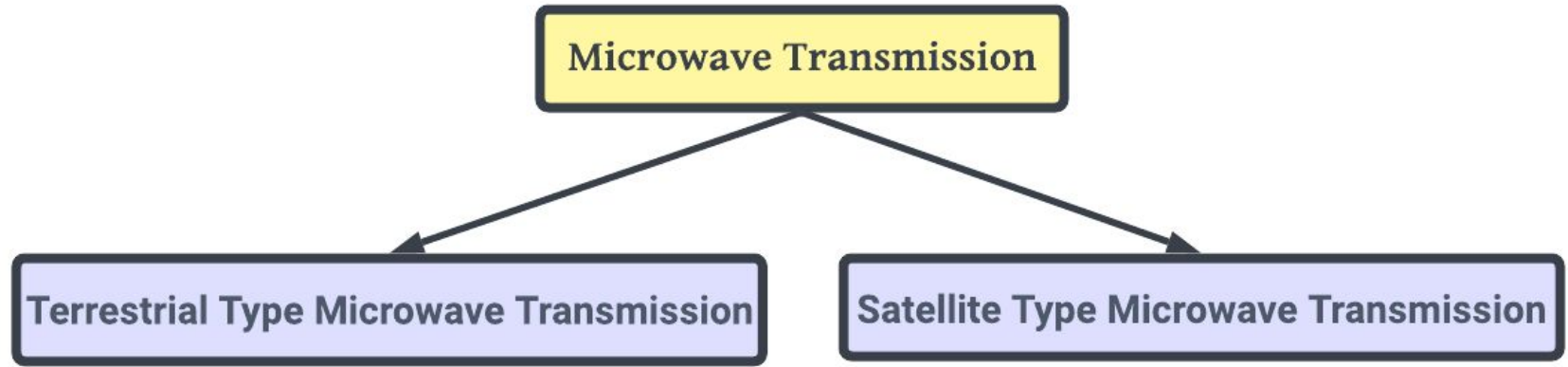
Components:

- **Transmitter:** Generates and modulates the signal
- **Receiver:** Receives and demodulates the signal

6.1.2. Transmission Media: Unguided Media(Wireless): Radio Waves



6.1.2. Transmission Media: Unguided Media(Wireless): Microwave



6.1.2 Unguided Media(Wireless):Terrestrial Microwave

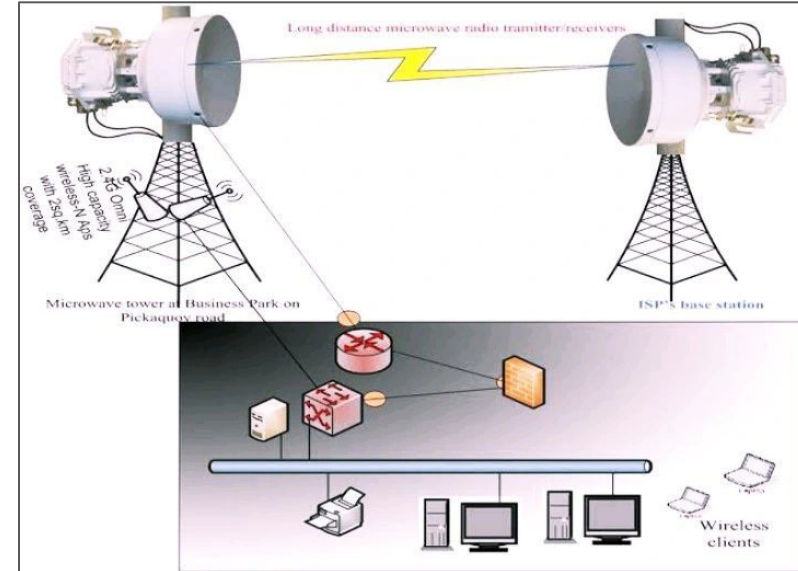
Property	Detail
Frequency Range	1 GHz - 300 GHz
Propagation	Unidirectional (line-of-sight)
Antenna	Parabolic dish antennas
Range	~50 km between towers

Characteristics:

- Requires line-of-sight (LOS) — no obstacles between sender & receiver
- Repeater towers placed at intervals on hills/tall buildings
- Higher frequency = more data capacity

Applications:

- Mobile phone networks (backbone)
- Long-distance telephone calls
- Cable TV distribution



6.1.2 Unguided Media(Wireless):Satellite Microwave

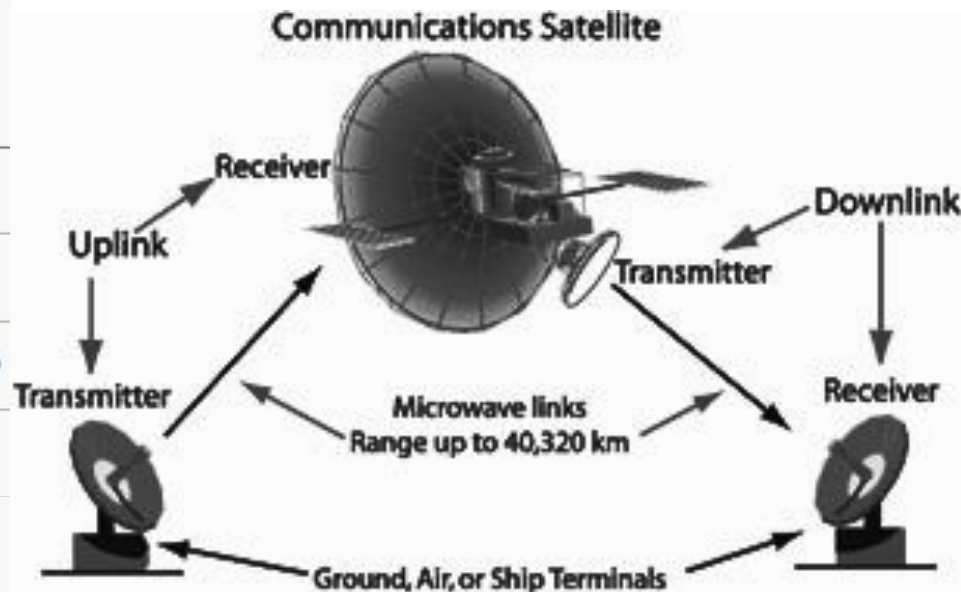
B) Satellite Microwave

Property	Detail
Orbit	GEO (36,000 km), MEO, LEO
Propagation	Very wide coverage area
Delay	High latency (especially GEO ~270ms)
Frequency	1-10 GHz (uplink/downlink)

Characteristics:

- Satellite acts as a **repeater in space**
- Uplink: Ground → Satellite | Downlink: Satellite → Ground
- Wide area coverage (entire continents)

Applications: GPS, Direct-to-home (DTH) TV, Internet in remote areas, Weather forecasting



6.1.2 Unguided Media(Wireless):Infrared

Property	Detail
Frequency Range	300 GHz - 400 THz
Range	Very short (few meters)
Propagation	Line-of-sight, cannot penetrate walls
Security	High (confined to room)

Characteristics:

- Cannot pass through solid objects
- No government regulation/licensing needed
- Cannot be used outdoors (sunlight interference)

Applications:

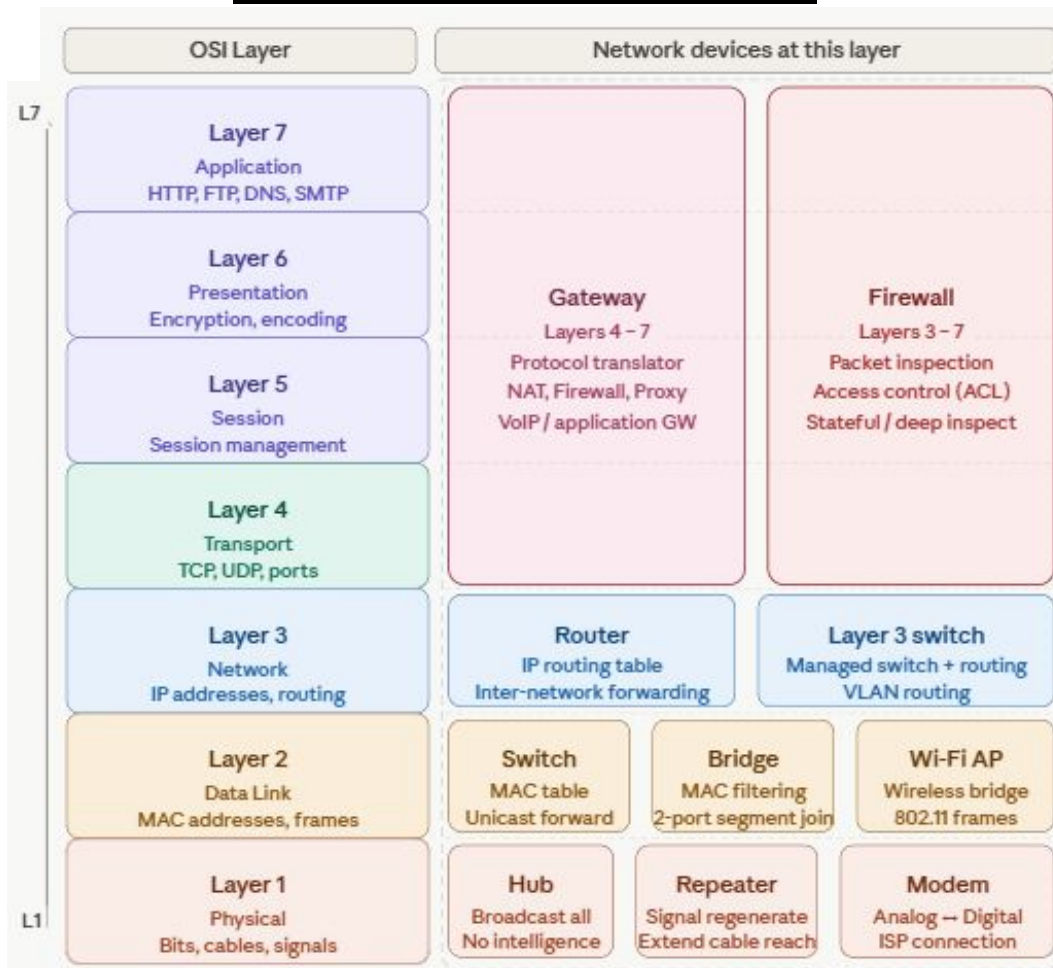
- TV/AC Remote controls
- IrDA (Infrared Data Association) communication
- Short-range device communication



6.1.2. Transmission Media: Unguided Media (Wireless)

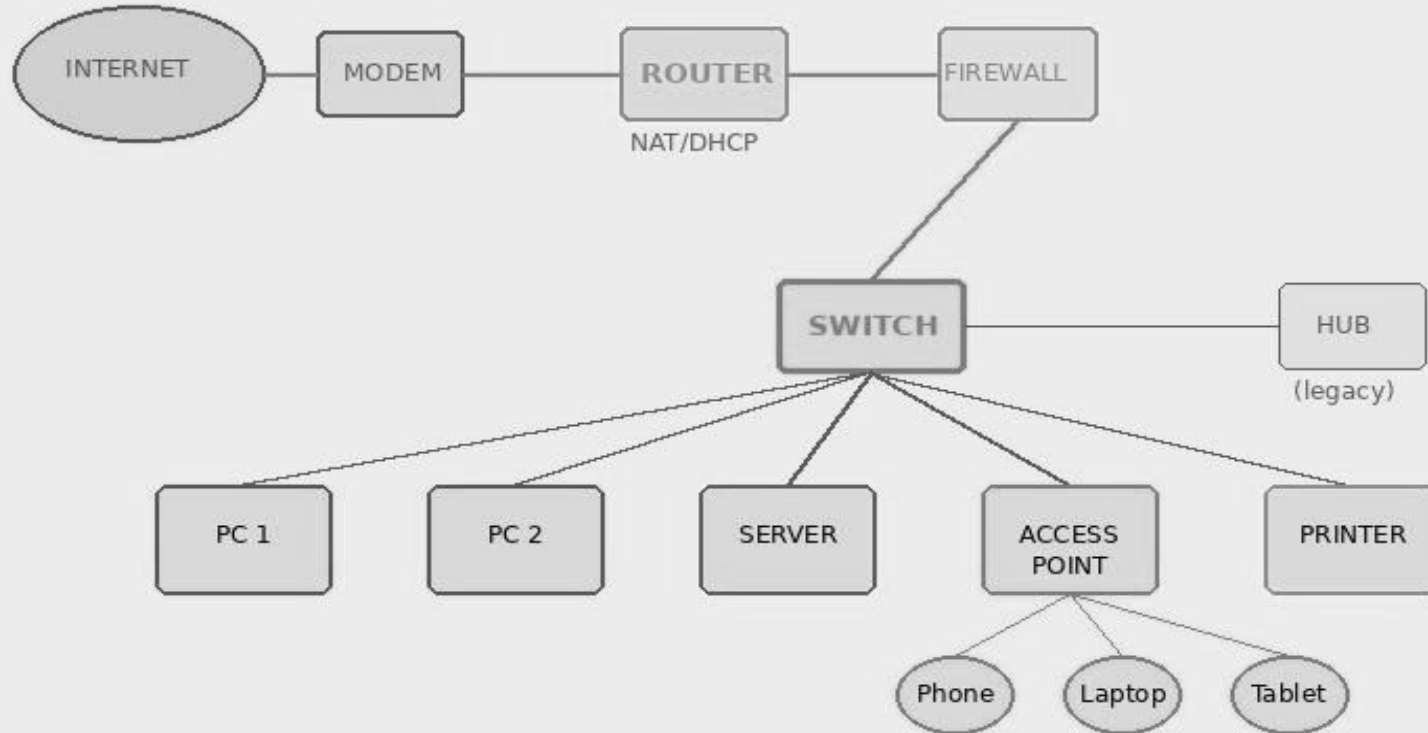
Property	Radio waves 3 KHz – 1 GHz	Microwaves 1 GHz – 300 GHz	Infrared 300 GHz – 400 THz
Range	Very long	Medium – Long	Very short (meters)
Direction	Omnidirectional	Unidirectional	Line-of-sight
Penetration	High (walls, buildings)	Medium	None
Interference	High	Medium	Low
Cost	Low	Medium – High	Low
Example use	FM/AM Radio TV, WI-FI	Mobile backbone GPS, Satellite TV	TV Remote IrDA, AC remote

6.2. Network Devices



6.2. Network Devices

Network Devices in a Typical Network Setup



6.2. Network Devices

1. Network Interface Card (NIC)

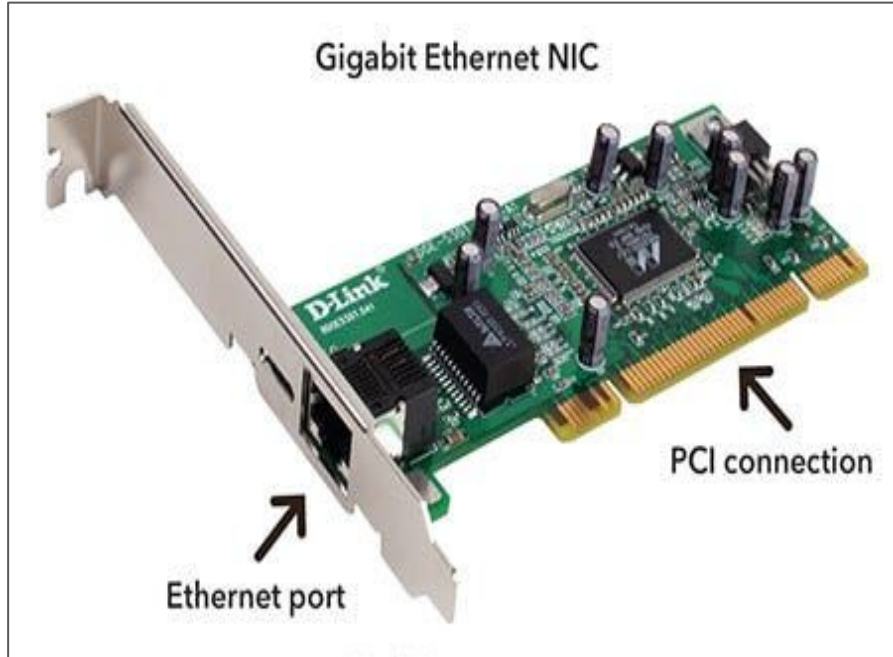
- ❑ A **NIC (Network Interface Card)** is a **hardware component** that connects a device (computer, laptop, printer, etc.) to a network.
 - Provides a physical connection to the network
 - Stores the device's MAC address

- ❑ **MAC Address:** A MAC Address is a unique hardware identifier assigned to a NIC by the manufacturer. It is used for identifying devices on a local network (LAN). It is a 48-bit address (6 bytes), written in hexadecimal format. Example: 00:1A:2B:3C:4D:5E

- ❑ **Types of NIC:**
 1. **Wired NIC:** Uses Ethernet cable. Example: Desktop LAN port
 2. **Wireless NIC:** Uses Wi-Fi signals. Example: Laptop Wi-Fi adapter

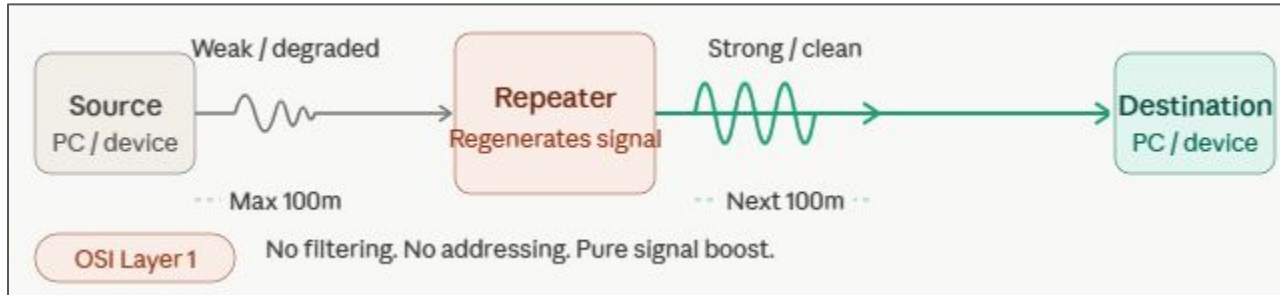
6.2. Network Devices

1. Network Interface Card (NIC)



6.2. Network Devices

2. Repeater — Layer 1 (Physical)



- A repeater operates purely at the physical layer.
- It receives a degraded electrical (or optical) signal, amplifies and regenerates it to its original shape and strength, then retransmits it.
- It has zero intelligence — it cannot read MAC addresses or IP addresses, cannot filter traffic, and simply extends the cable's effective reach beyond its physical limit (~100m for Ethernet).



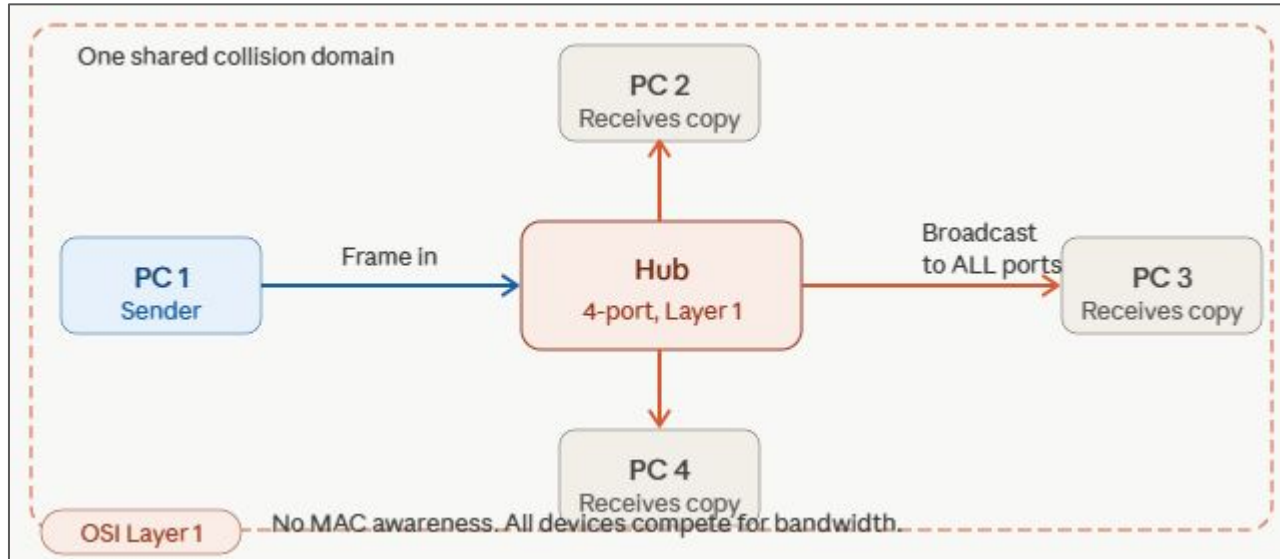
6.2. Network Devices

3. HUB— Layer 1 (Physical)

- A hub is essentially a multi-port repeater. When a frame arrives on any one port, the hub immediately floods it out of every other port — regardless of who the intended recipient is.
- All connected devices share a single collision domain, meaning only one device can transmit at a time. If two transmit simultaneously, a collision occurs and both must retransmit.
- Hubs are obsolete today but important to understand as the predecessor to the switch.

6.2. Network Devices

3. HUB— Layer 1 (Physical)



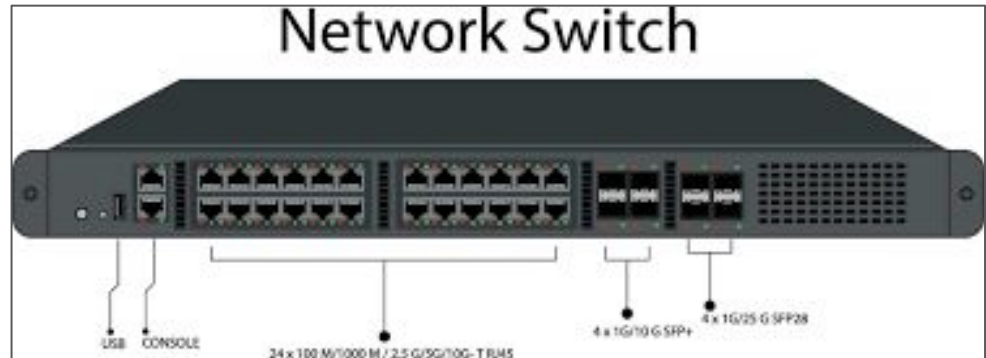
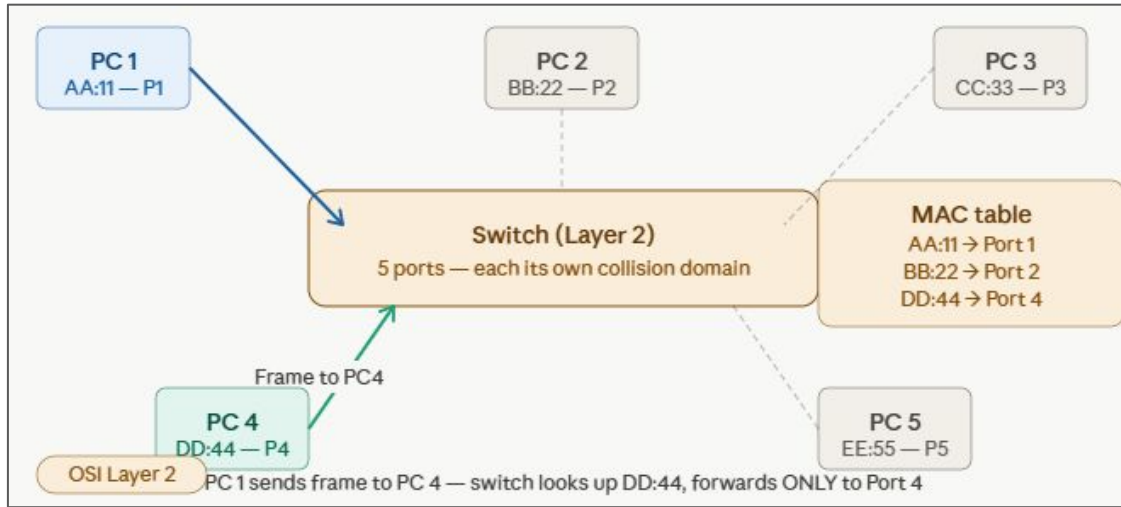
6.2. Network Devices

4. Switch— Layer 2 (Data Link)

- ❑ A switch is a multi-port device with a MAC address table for every port.
- ❑ When PC 1 sends a frame to PC 4, the switch looks up the destination MAC address in its table and forwards the frame only to the correct port — no other device sees it.
- ❑ This creates a separate collision domain per port, dramatically reducing collisions and improving throughput compared to a hub.
- ❑ Modern managed switches also support VLANs, QoS, and even Layer 3 routing.

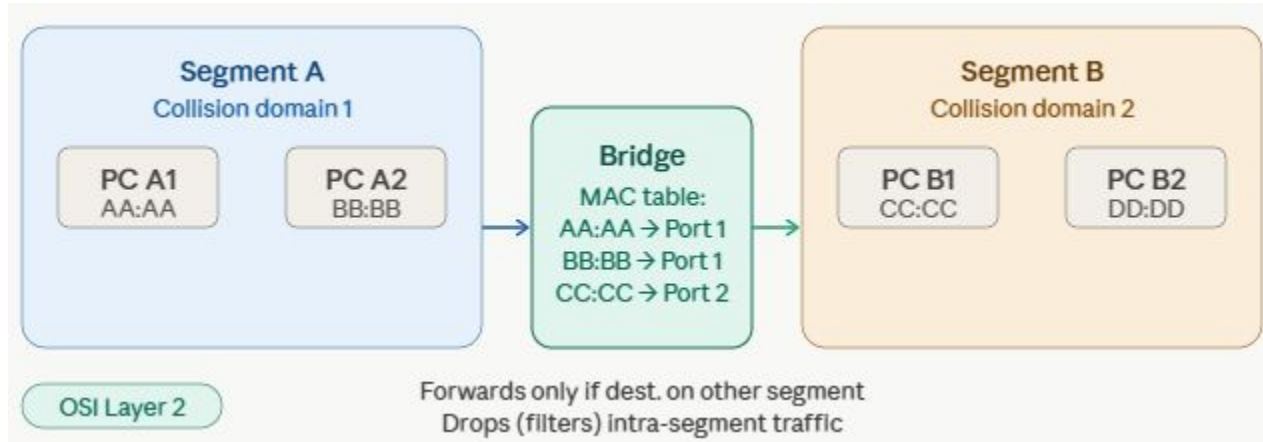
6.2. Network Devices

4. Switch— Layer 2 (Data Link)



6.2. Network Devices

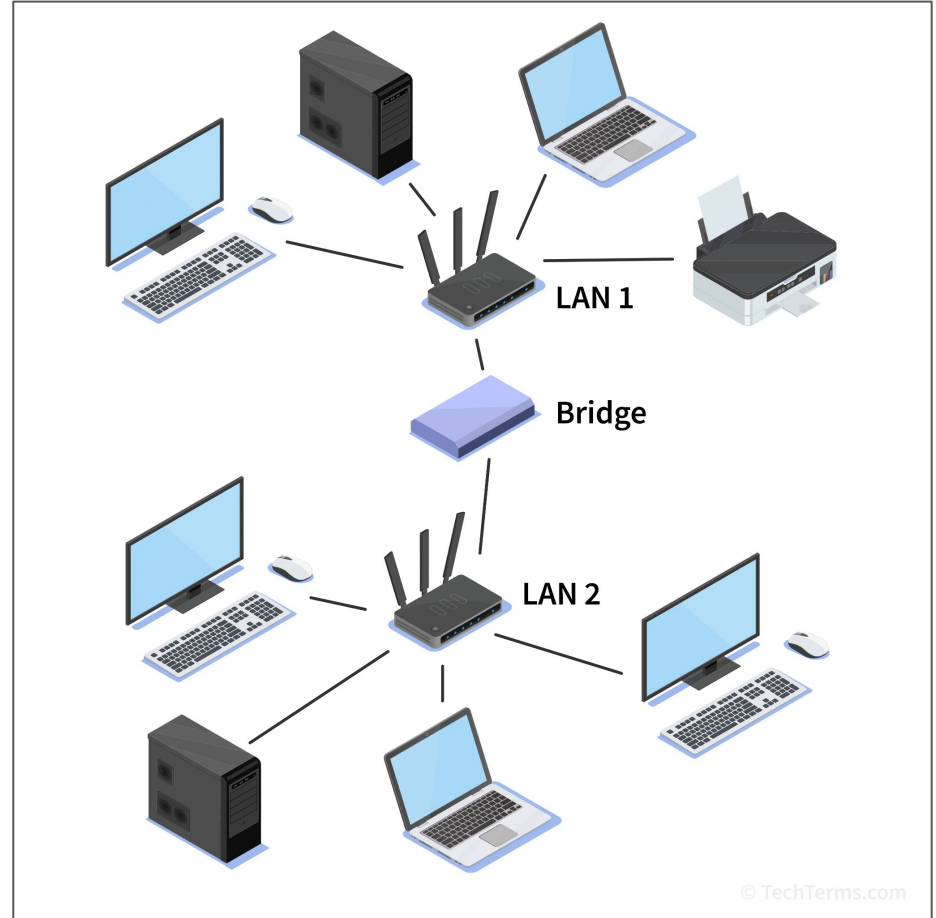
5. Bridge— Layer 2 (Data Link)



- ❑ A bridge connects two network segments and learns which MAC addresses are on which side.
- ❑ If a frame's destination is on the same segment it came from, the bridge drops it (filters). Only if the destination is on the other segment does it forward the frame.
- ❑ This separates two collision domains, reducing congestion significantly compared to a hub.

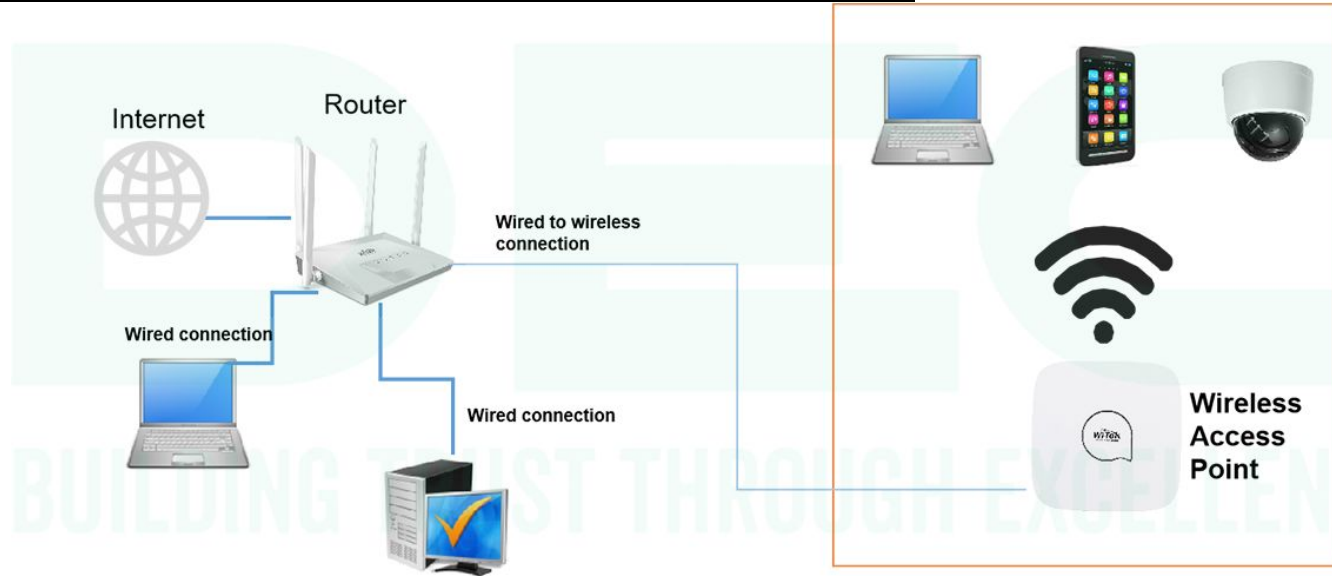
6.2. Network Devices

5. Bridge— Layer 2 (Data Link)



6.2. Network Devices

6. Access Point (AP)— Layer 2 (Data Link)

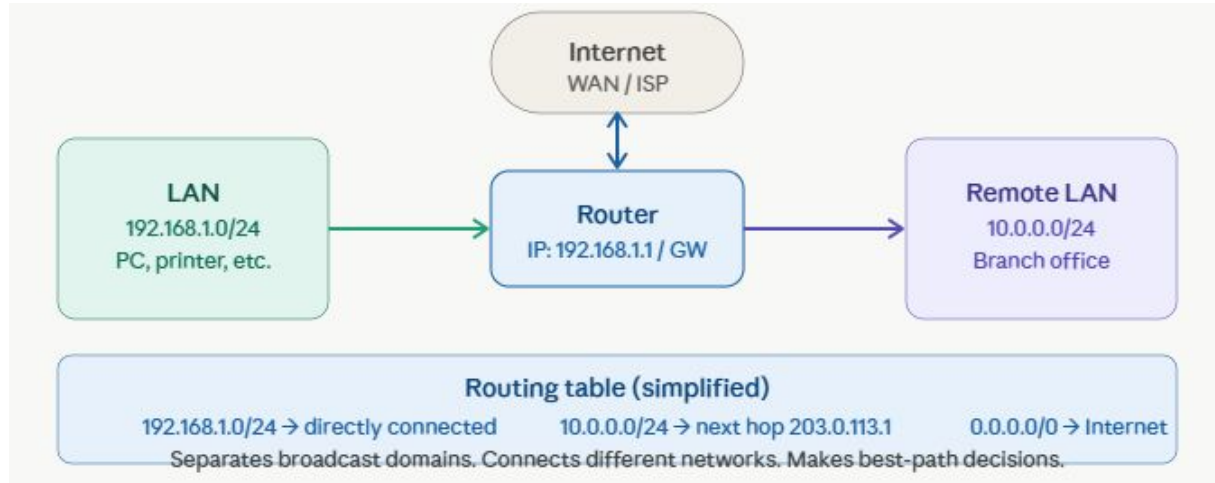


It is a networking hardware device that allows Wi-Fi-enabled devices, such as laptops and smartphones, to connect to a wired network.

- Acts as a bridge between wired Ethernet and wireless Wi-Fi devices.
- An AP only provides Wi-Fi signal and it does not route traffic or assign IP addresses.

6.2. Network Devices

7. Router — Layer 3 (Network)

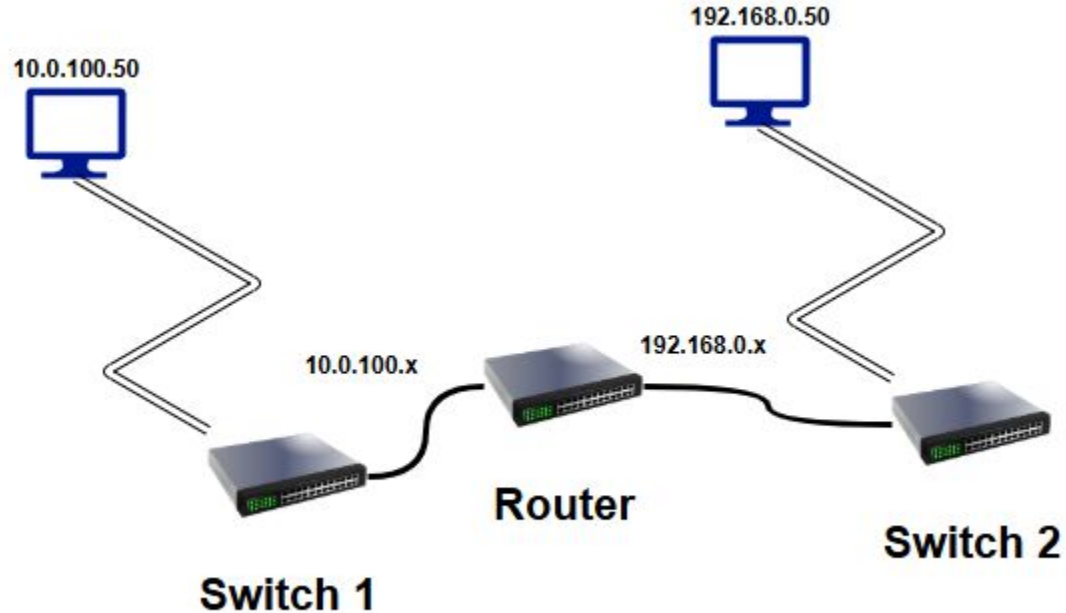


It is a networking device that connects multiple networks and directs data packets between them using IP addresses.

- It uses IP Addresses to determine the best path for data packets to travel.
- It maintains a Routing Table to make these decisions.
- Routers stop broadcast traffic, isolating networks from each other.

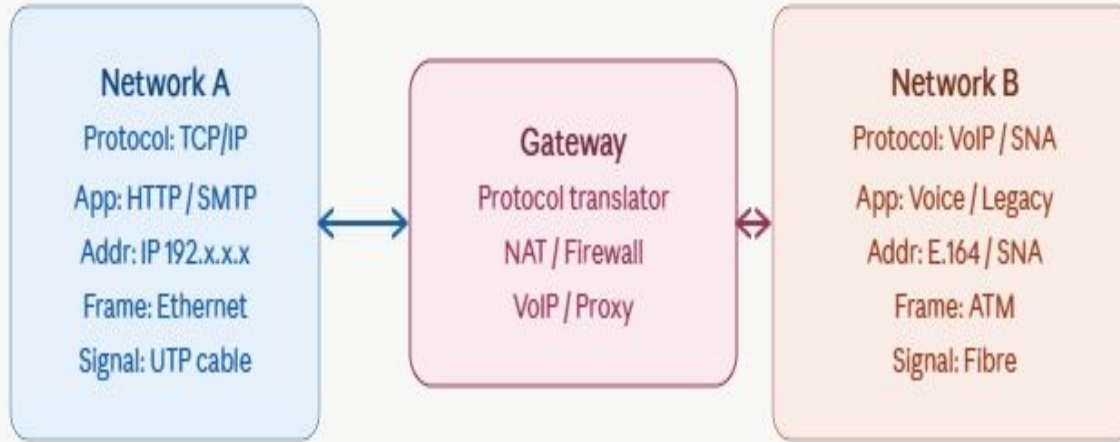
6.2. Network Devices

7. Router — Layer 3 (Network)

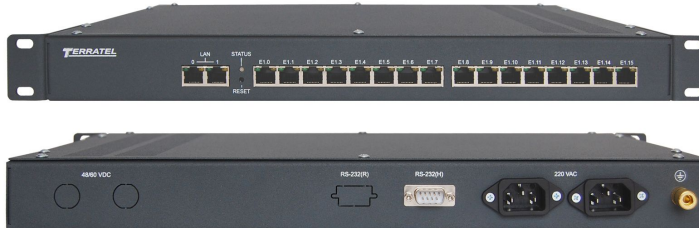


6.2. Network Devices

8. Gateway— Layer 4-7 (All Upper Layers)



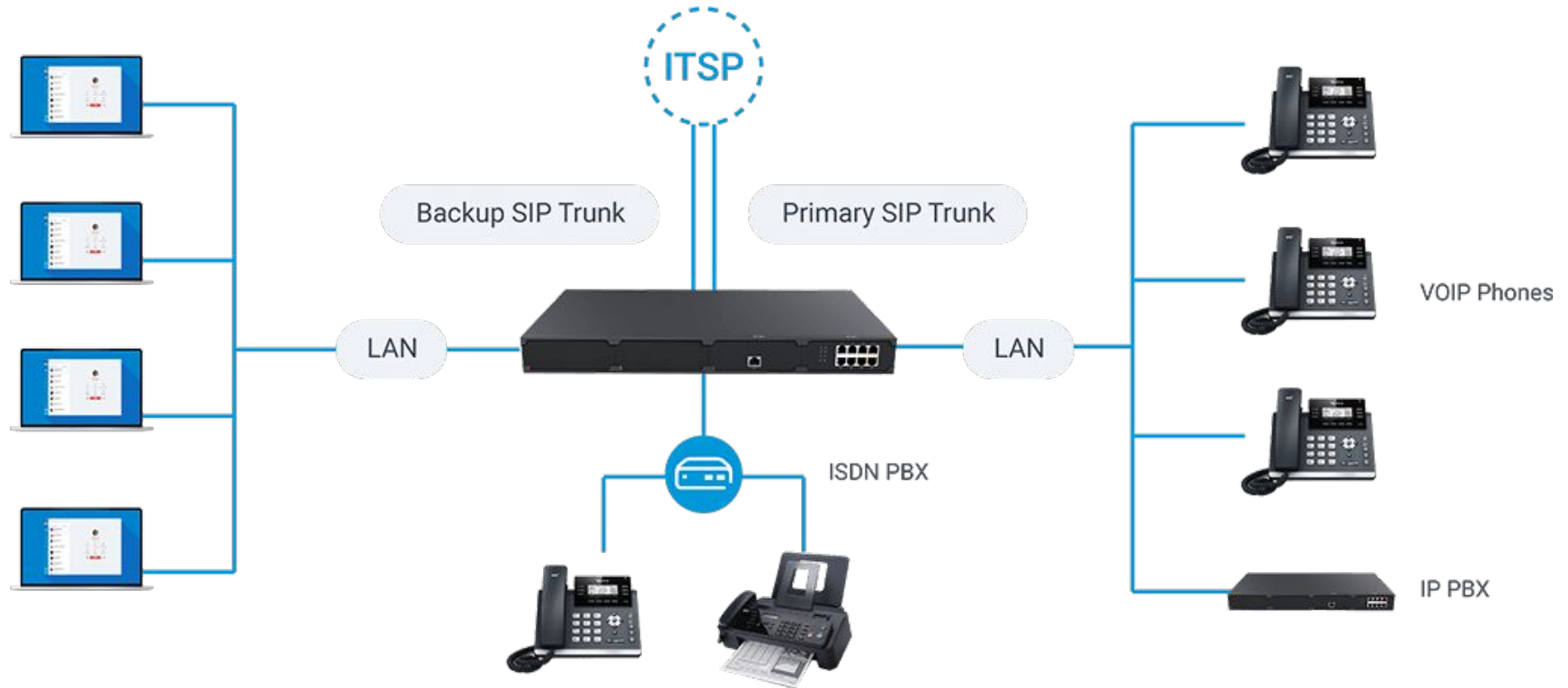
OSI Layers 4-7 Understands and translates entire protocol stacks — most intelligent network device



A gateway is the most intelligent network device. It can translate between completely different protocol architectures — for example, converting TCP/IP traffic to VoIP signalling, or bridging an Ethernet LAN to a legacy IBM SNA network. Your home router acts as your network's default gateway. Enterprise gateways also perform NAT (Network Address Translation), firewall inspection, and proxy services.

6.2. Network Devices

8. Gateway— Layer 4-7 (All Upper Layers)



6.2. Network Devices

9. Firewall— Layer 4-7 (All Upper Layers)

A firewall is a network security system—either hardware, software, or both - that monitors and controls incoming and outgoing traffic based on predetermined security rules. It establishes a barrier between a trusted internal network and an untrusted external network (like the Internet).



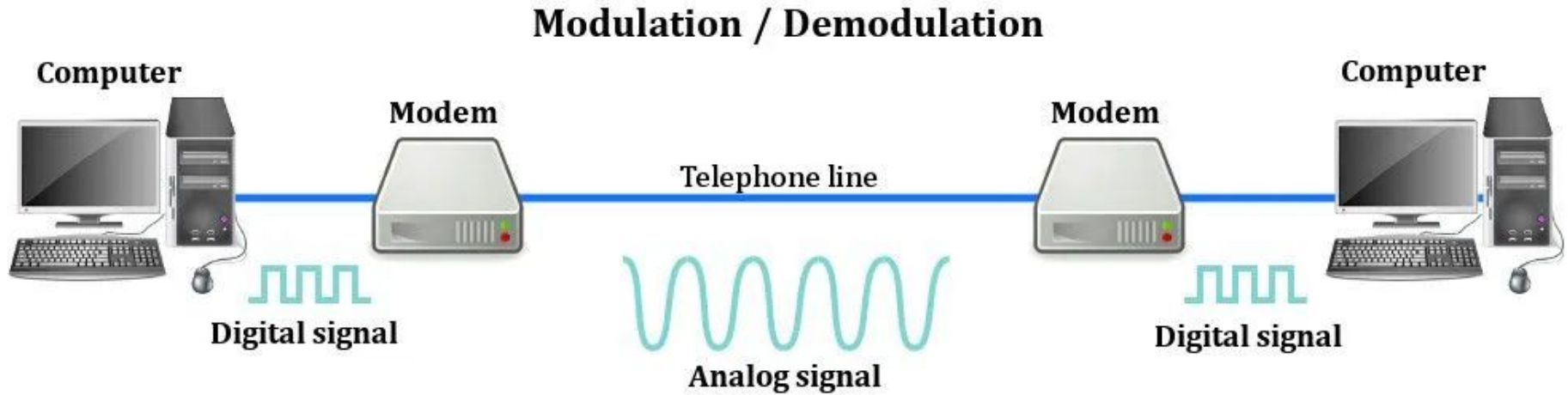
6.2. Network Devices

10. Modem (Modulator-Demodulator)— Layer 2 (Data Link)

- ❑ A **modem** (short for **modulator–demodulator**) is a networking device that enables communication between a computer (digital system) and a transmission medium that carries **analog signals**, such as telephone lines or cable systems.
- ❑ Computers work with digital signals, but many communication channels (especially older systems like telephone networks) transmit analog signals. A modem acts as a translator between these two formats.
- ❑ The modem performs two essential operations:
 - ❑ **Modulation:** Converts digital data (0s and 1s) from a computer into analog signals. These analog signals can travel over media like telephone lines
 - ❑ **Demodulation:** Converts incoming analog signals back into digital data. This allows the receiving computer to understand the information

6.2. Network Devices

10. Modem (Modulator-Demodulator)— Layer 2 (Data Link)



6.2. Network Devices

Feature	Switch	Router
Primary Role	Connects devices within the same network (creates a LAN).	Connects different networks together (routes between LANs/WANs).
OSI Layer	Data Link (Layer 2) — some can also operate at Layer 3.	Network (Layer 3).
Addressing Used	MAC address (e.g., 00:1A:2B:3C:4D:5E)	IP address (e.g., 192.168.1.1)
Decision Logic	Forwards frames based on a MAC address table (CAM table).	Forwards packets based on a routing table (static routes, OSPF, BGP).

Feature	Switch	Router
Communication Scope	Inside a single broadcast domain (one subnet). Creates only one broadcast domain unless VLANs are used.	Separates broadcast domains. Each router interface defines a new subnet.
Key Functions	Learning MAC addresses, forwarding/filtering frames, VLAN segmentation.	Path selection, NAT, firewall, DHCP server, traffic prioritization (QoS).
Hardware Interfaces	Multiple RJ45/SFP ports; all in the same subnet.	Separate physical ports for internal (LAN) and external (WAN) networks.
Typical Location	Inside a building, connecting servers, PCs, access points.	At the edge of your network (connecting to a modem/ISP), or enterprise core routing between VLANs.

7. Network Topology

Network topology is the arrangement of devices (nodes) and connections (links) in a computer network. It shows how computers, servers, and other network devices are connected and how data flows between them. Choosing the right topology is important because it affects the performance, cost, reliability, and security of the network. There are two main types of topology:

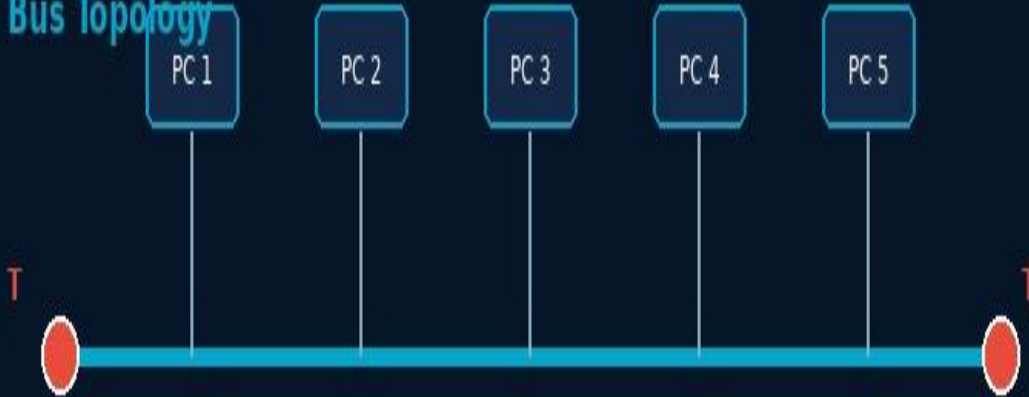
- **Physical Topology:** The actual physical layout of cables and devices.
- **Logical Topology:** How data moves across the network, regardless of physical layout.

All the following topologies (Bus, Star, Mesh, Ring, Tree, Hybrid) are physical topologies. Logical topology refers to how data flows in the network, such as CSMA/CD in Ethernet or token passing in ring networks.

1. Bus Topology

All devices share a single common cable (backbone)

Bus Topology



All devices share a single backbone cable. Terminators (T) prevent signal reflection.

Bus Topology

Definition:

All devices connect to a single central cable (backbone). Data is broadcast to all nodes; only the intended recipient accepts it. Terminators at each end prevent signal reflection.

✓ Advantages:

Simple and inexpensive to set up. Easy to add new devices. Requires less cable than star.

✗ Disadvantages:

A break anywhere in the backbone brings down the entire network. Difficult to troubleshoot. Performance degrades with many devices.

📌 Real Uses:

Early 10Base2 Ethernet (Thinnet), cable TV distribution, older office networks.

2. Star Topology

All devices connect to a central hub or switch

Star Topology



All devices connect to a central Hub/Switch.
Easy to add/remove. Hub failure = network failure.

Star Topology

Definition:

Every device has a dedicated point-to-point connection to a central hub or switch. Data passes through the central device. The most widely used topology today.

✓ Advantages:

Easy to add/remove devices. Fault isolation: one device failure doesn't affect others. Easy to troubleshoot and manage.

✗ Disadvantages:

If the central hub/switch fails, the entire network goes down. Requires more cabling than bus. Central device is a single point of failure.

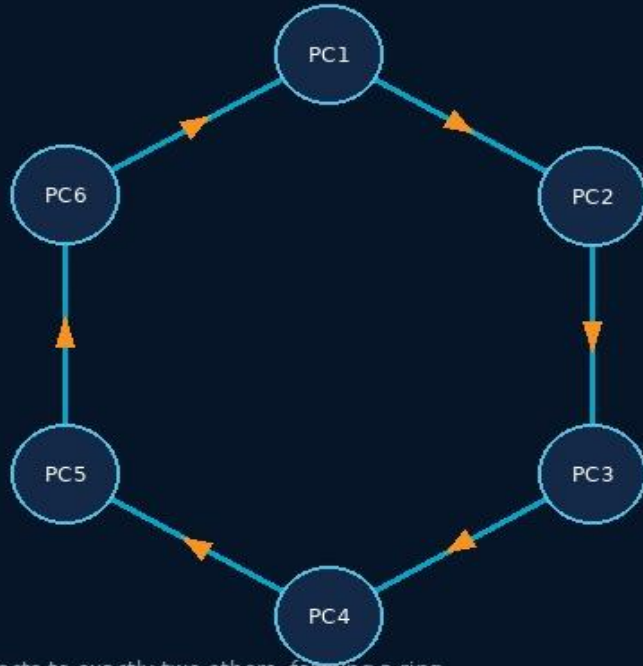
📌 Real Uses:

Modern home networks, office LANs, school computer labs — virtually all modern Ethernet networks.

3. Ring Topology

Each device connects to exactly two neighbors, forming a closed loop

Ring Topology



Each device connects to exactly two others, forming a ring.
Token passing used. One break = network failure.

Ring Topology

Definition:

Devices are connected in a circular loop. Data travels in one direction (unidirectional) or both directions (bidirectional/SONET). Token passing controls access — only the device with the 'token' can transmit.

✓ Advantages:

Equal access for all devices. Predictable performance under load. No data collisions (token passing). Simple to install.

✗ Disadvantages:

One cable break can take down the entire network (single ring). Adding/removing devices disrupts the network. Slower than star with many nodes.

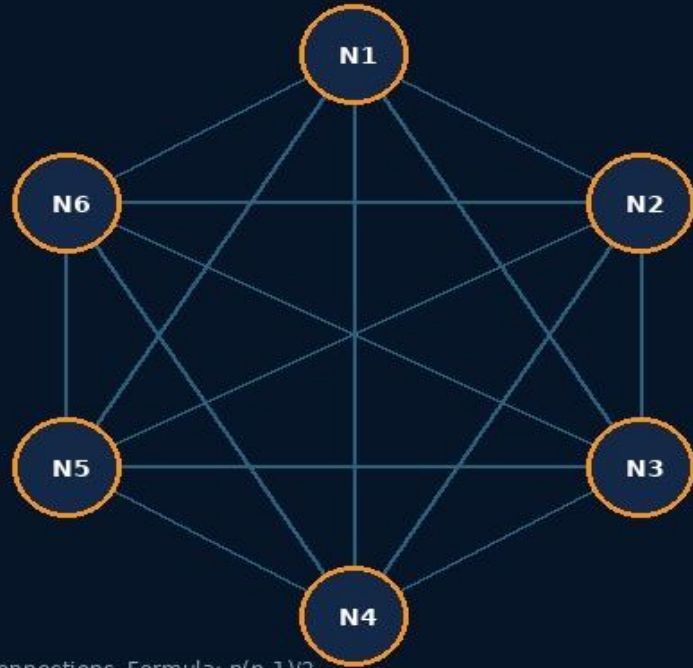
📌 Real Uses:

IBM Token Ring networks, SONET/SDH fiber rings in telecom, metropolitan fiber networks.

4. Mesh Topology

Every device has a direct connection to every other device

Mesh Topology (Full Mesh)



6 nodes $\times$ 15 connections. Formula: $n(n-1)/2$
Highly fault tolerant. Very expensive to implement.

Mesh Topology

Definition:

In a full mesh, each node is directly connected to every other node. In a partial mesh, only some nodes have redundant connections. Number of links in full mesh: $n(n-1)/2$. E.g., 6 nodes = 15 links.

✓ Advantages:

Extremely fault tolerant — multiple paths exist. If one link fails, data reroutes automatically. Highest security and reliability. Traffic shared across multiple paths.

✗ Disadvantages:

Very expensive — $n(n-1)/2$ cables needed. Complex installation and maintenance. Each device needs multiple network ports.

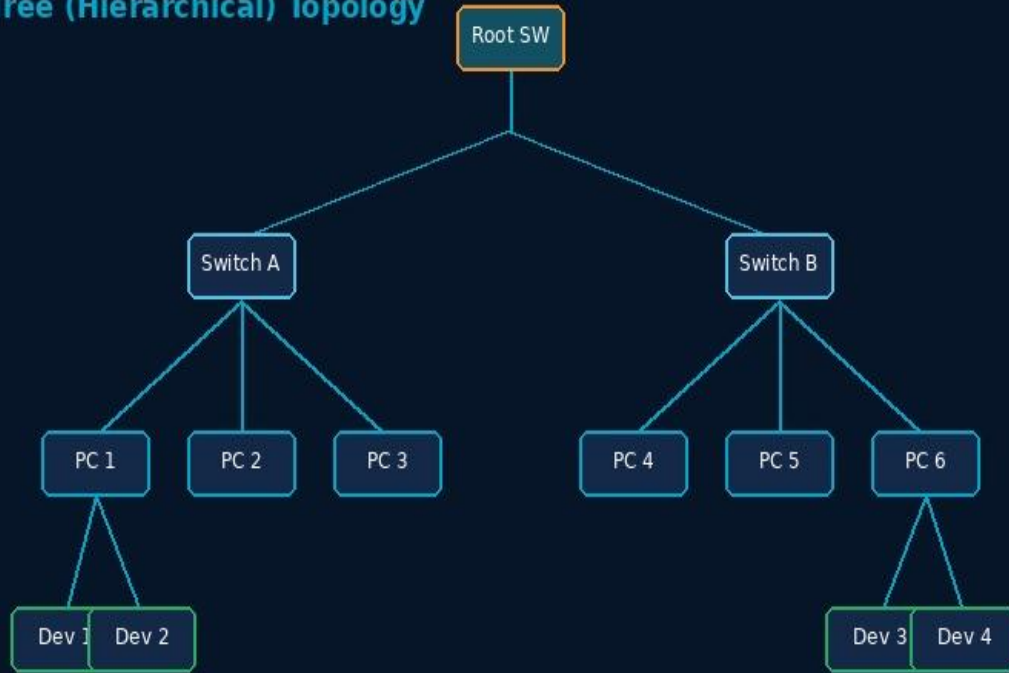
📌 Real Uses:

Internet backbone routers, military networks, critical infrastructure (power grids), data center interconnects.

5. Tree Topology

Hierarchical combination of Star and Bus topologies

Tree (Hierarchical) Topology



Combines Star & Bus. Hierarchical levels. Root failure = total failure.

Tree Topology

Definition:

Devices are arranged in a hierarchy with a root switch at the top, branching down through intermediate switches to leaf nodes (PCs). Combines the star structure at each level with bus-like trunk connections.

✓ Advantages:

Highly scalable — add branches easily. Easy to segment and manage. Supports large networks with clear hierarchy.

✗ Disadvantages:

Root failure affects all branches. More cabling required than bus. Backbone failure isolates sub-trees.

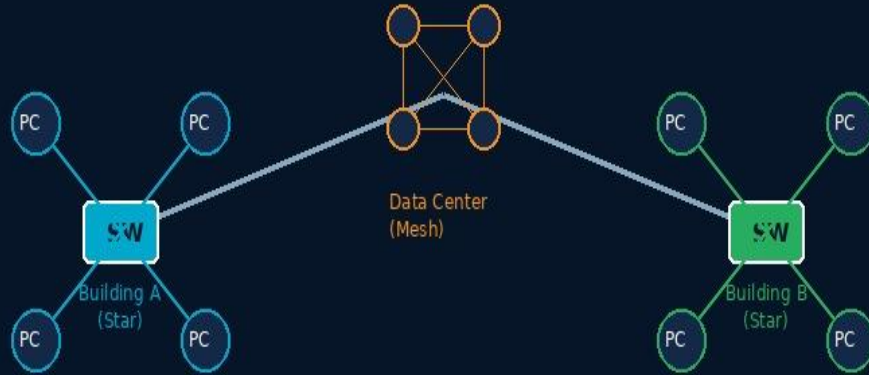
📌 Real Uses:

Corporate networks (headquarters + branches), university campus networks, ISP distribution networks.

6. Hybrid Topology

Combination of two or more topology types

Hybrid Topology (Star-Bus-Mesh)



Star segments: Easy management per building
Bus backbone: Connects buildings efficiently
Mesh core: Redundant paths for critical servers

Hybrid Topology

Definition:

A hybrid topology combines two or more different topologies to leverage the benefits of each. A common example is Star-Bus: multiple star-wired buildings connected by a bus backbone.

✓ Advantages:

Flexible design for complex needs. Fault isolation (one topology failure doesn't affect others). Performance optimized per segment. Highly scalable.

✗ Disadvantages:

Complex to design and maintain. Higher cost. Troubleshooting requires understanding multiple topology types.

📌 Real Uses:

Large enterprise networks, university campuses, data center networks, city-wide networks.

8. IP Address

An IP address(Internet Protocol address) a unique numerical label assigned to every device on a network that uses the Internet Protocol. The most common version is IPv4(32 bits), written in dotted-decimal notation—four numbers (0-255) separated by dots. Example: 192.168.1.10

It serves two main functions:

1. **Identification:** Uniquely identifies a host (computer, server, phone) or network interface.
2. **Location:** Provides a logical address that routers use to determine the path to the destination.

Think of it like a postal address:

- MAC address = Your name, useful only within your household (local network).
- IP address = Your full street address, city, and country, enabling anyone in the world to send you a letter (packet).

END
of
Chapter 6